DSAP small project about Noise Reduction

1. Spectral subtraction (SS): basic and multiband (MB)

You are going to experiment with different versions of the spectral subtraction (SS) speech enhancement algorithm.

First, you have to spend time tuning a basic SS algorithm, by playing with the two parameters (oversubtraction and flooring). Take care with notation, as the parameter alpha has a different meaning in my slides. The script is in MATLAB_code/spectral_subtractive/specsub.m.

You can then work with the SNR-dependent oversubtraction factor, which is explained in Berouti et al paper (see the code) and avoids tuning oversubtraction, so only flooring needs to be tuned.

Also, try with and without updating the noise spectrum estimate by activating and deactivating the simple VAD included in the code, and, for the case of deactivating the VAD, playing with the number of initial frames to estimate the noise spectrum (looking first at the signals to see which silence length there is at the beginning of the files, so you can use more than 5, the number by default in the code).

In case you have time enough, you can also compare results with the multiband SS algorithm (mband.m). Here you have also several parameters to play with: number of bands, frequency spacing, etc. Study the subject in the paper given in the .m file.

2. Noise spectrum estimation (NSE) with spectral subtraction (SS)

You are going to experiment with noise spectrum estimation (NSE) techniques in the framework of the spectral subtraction (SS) speech enhancement algorithm. The script is MATLAB_code/noise_estimation/specsub_ns.m.

As you can see in the Readme file (in the same folder), there up to 7 different noise estimation techniques implemented. You have to compare two of them, which can be chosen by yourself, perhaps reading some paper that makes comparisons. If you asked me a recommendation, I would say that you choose the ones called Martin's and mcra2. You will have to present the chosen techniques in class pointing out the differences between them.

Indeed, you have to know well the spectral subtraction (SS) algorithm, already explained quickly in class (oversubtraction and flooring). Take care with notation, as the parameter alpha has a different meaning in my slides. The SS implementation is based on a SNR-dependent oversubtraction factor, which is explained in Berouti et al paper [1] and avoids tuning oversubtraction, so only flooring needs to be tuned, which might be different for each noise estimation algorithm. Each NSE algorithm also have parameters that have to be tuned.

When presenting the algorithms in class, you will not need to present SS itself as there will be a team presenting it before yours.

3. Wiener filter (WF) and voice activity detection (VAD)

The speech enhancement approach assigned to your team is based on the Wiener filter explained in class. But you will work with the script MATLAB_code/statistical_based/wiener_as.m. This means that you have to establish the relationship between two different notations by comparing the one in class with the one in the code (which uses a priori and a posteriori SNRs), checking that they correspond to the same mathematical expressions. The paper from Scalart et al mentioned in the code can help with that, though the notation differs; also, the Hu-Loizou paper from 2004, mentioned in the Readme file may help you. When presenting the technique in class, use this notation in the code.

By tuning the free parameters (e.g. a_dd) you have to obtain the best results with the Wiener algorithm. First do it without VAD, so the noise spectrum is only estimated from the beginning of the file and not updated along the file. Then include the VAD implemented in the given code. Those results will be compared in class with the ones from Team4. Please, in the submission include results with different values of the parameters.

You can find a presentation of that VAD in the paper from Sohn et al. You should understand well that VAD approach, both the mathematical expressions and the code implementation.

4. Wiener filter (WF) with multi-tapered wavelet thresholding (WT)

The speech enhancement approach assigned to your team is based on the Wiener filter (WF) and includes noise spectrum estimation with the wavelet-thresholding (WT) multi-taper spectra. The plain WF technique itself is also included in your work.
The corresponding Matlab function is wiener_wt.m, that you can find in Matlab_code/statistical_based. Also, wiener_as.m for the plain WF.

You will need to know/study:
- The Wiener filter. Comments for Team3, first paragraph, are also valid for you.
- Wavelets and wavelet thresholding with multi-taper spectra.
There are papers referenced within the code. In the presentation, spend more time with wavelets than with the multi-taper spectrum, since it is a spectral estimation technique not very different from the Welch technique used in class (it is also called, more explicitely, multiwindow spectra)

By tuning the free parameters (e.g. a_dd) you have to obtain the best results with the Wiener filter technique. First do it without VAD, so the noise spectrum is only estimated from the beginning of the file and not updated along the file. Then include the VAD implemented in the given code. Those results will be compared in class with the ones from Team3.

After that, you have to tune parameters and test the wt-based technique, trying to improve with it the results from the plain Wiener filter. For that, you can play with the parameters of the wavelet decomposition.

You do not need to understand the rationale of the VAD algorithm, it will be done by Team3, who will explain it in class. The wavelet decomposition it's quite a wide subject, if you do not know it look only at the basic ideas.

5. logSMME with SPU:

The speech enhancement approach assigned to your team, logMMSE, is based on minimization of a mean-square error, like the Wiener filter (WF) and includes speech-presence uncertainty (SPU) estimation.

You will have to read a few papers about logMMSE and the SPU techniques implemented in the script logmmse_SPU.m, which you have to use and understand well. (It is in folder MATLAB_code/statistical_based). Alternatively you can study it with the Loizou's book.

The logMMSE technique is explained e.g. in the original paper from Ephraim and Malah. Understand the basic assumptions of it. Don't forget to relate this method with the Wiener filter.

You do not need to understand the rationale of the VAD algorithm, it will be done by Team3, who will explain it in class.

To avoid having too many results in the final comparison in class, you may only include results from two SPU options in the Table of Results.

Make sure you try different values for the parameters included in the main WF loop and also in the various SPU estimation techniques.

6. Subspace-based KLT and perceptually-motivated KLT

The speech enhancement technique assigned to your team is based on the subspace based alternative approach. First, the generalized subspace algorithm with embedded pre-whitening, which includes multi-taper (taper means window here) covariance matrix estimation, as you can see in the script klt.m that implements the whole technique. You can find it in Matlab_code/subspace folder.

To understand well the papers that are given inside the code, you may have to study some basic concepts and techniques concerning matrix algebra and/or others.

Try to get the best results with the technique by tuning parameters or using algorithm variants. In the presentation try to focus on the basic points and minimizing the use of math expressions.

7. Audible noise supression technique and perceptually meaningful quality measures

This topic has two parts which are related up to some extend. First, the audible noise suppression algorithm (audnoise.m in folder MATLAB_code/statistical_based) has to be understood (see the reference paper given inside the code) and tested. It includes a decomposition in critical bands, where the noise is estimated.

Although a VAD algorithm is included in the code, you do not need to understand the rationale of it, as it will be done by Team3, who will explain it in class. For testing, different values for the parameters must be used, looking for the best results with our data.

The second part is also related to perceptual models. It consists of studying the foundations and implementation of the various quality measures that all the teams will use to evaluate the performance of their algorithms, and especially the standardized PESQ measure.

The testing work of this second part will consist of comparing the various measures regarding the quality of the noisy (unprocessed) signals, and with respect to 2 variables: SNR and type of noise. For extracting conclusions from the table of results you can also plot the measures along time in a frame-by-frame basis to check their behaviour with respect to speech, silence, time dynamics, etc

Reference for all the works:

Philipos C. Loizou, “Speech Enhancement. Theory and Practice”, 2nd Ed., CRC Press, 2013.